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D.K.T.E Society's Textile and Engineering Institute, Ichalkaranji

STEAMROLLER

Synopsis of Proposed Work Submitted to

Department of CSE(AIML) Engineering

For the Degree of

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In
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by

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▪ INTRODUCTION

➢ Domain

Speech processing and Automatic Speech Recognition (ASR) are key domains in Artificial Intelligence and signal processing, enabling machines to understand and respond to human speech. These technologies form the foundation of modern voice-based systems such as virtual assistants, speech-to-text applications, and hands-free communication systems.

➢ Problems

Most existing ASR systems are designed and trained for fluent speech. They perform poorly when handling disfluent speech containing repetitions, prolongations, or speech blocks. This limitation significantly affects People Who Stutter (PWS), leading to inaccurate recognition, communication barriers, and reduced accessibility to voice-enabled technologies.

➢ Solutions

Advancements in deep learning, neural network–based ASR models, large language models (LLMs), and speech-to-text–text-to-speech (STT–TTS) pipelines provide potential solutions. Intelligent text repair and multi-stage speech processing approaches can improve recognition accuracy and enhance speech intelligibility.

➢ Narrowed Domain - Proposed Topic

The proposed STEAMROLLER system focuses on processing disfluent speech by integrating ASR with intelligent text repair and speech synthesis. The system aims to transform disfluent speech into fluent output while preserving the original meaning.

➢ Applications

The proposed approach is applicable in assistive communication systems, inclusive voice interfaces, telecommunication platforms, education tools, and accessibility-focused AI solutions for people with speech disfluencies.

▪ RELATED WORK

➢ Paper 1: Automated Dynamic Stuttered Speech Recognition System using MFCC and Enhanced Logistic Regression

- **Summary of Work:**

This paper proposes a stuttered speech recognition and conversion system using Weighted Mel Frequency Cepstral Coefficients (MFCC) for feature extraction and an enhanced Logistic Regression model for detecting repetitions and prolongations in stuttered speech. The system focuses on identifying stuttering events and converting recognized text into speech to improve communication fluency.

- **Limitations:**

1. Uses traditional machine learning models with limited capability to capture contextual and linguistic dependencies.

2. MFCC-based features do not fully represent complex speech variations.

3. Focuses mainly on detection rather than comprehensive disfluency repair.

4. Limited robustness for real-world conversational speech.

➢ Paper 2: Speech Disfluency Removal and Augmentation for Enhanced Voice Technology Interaction

- **How limitations of Paper 1 are tackled:**

Paper 2 addresses the limited contextual understanding of Paper 1 by adopting deep learning–based models such as fine-tuned Whisper ASR and transformer-based text reconstruction, enabling improved handling of complex and diverse disfluent speech patterns.

- **Summary of Work:**

This work presents an end-to-end pipeline that transforms disfluent speech into fluent and natural-sounding output. It integrates Whisper ASR for improved recognition accuracy, a hybrid disfluency detection module, a T5-based text reconstruction model for semantic repair, and zero-shot TTS for fluent speech generation while preserving speaker identity.

- **Limitations:**

1. Relies on a linear processing pipeline with limited flexibility.
2. Lacks collaborative or iterative reasoning mechanisms for speech repair.
3. Computational complexity restricts deployment on low-resource platforms.

➢ Paper 3 (STEAMROLLER)

• How limitations of Paper 2 are overcome:-

- Paper 2 uses a single, fixed end-to-end pipeline, whereas STEAMROLLER introduces a multi-agent architecture.

- Instead of one model making all decisions, STEAMROLLER assigns distinct responsibilities to specialized agents (ASR agent, disfluency detection agent, linguistic repair agent, semantic validation agent).

- Enables iterative correction rather than one-pass correction, improving handling of complex repetitions, long prolongations, and contextual errors.

- Improves semantic consistency, which is a limitation in transformer-only repair models.

- Provides better adaptability to unseen disfluency patterns compared to static pipelines.

• Summary of Work:

- STEAMROLLER is designed for repairing disfluent speech of People Who Stutter (PWS).
- Converts disfluent speech to text using ASR
- Repair repetitions and prolongations.
- Preserve speaker intent and meaning.
- Repaired text is converted back to fluent speech using TTS..

• Scope for Modification and Implementation:

- Multi-agent framework can be reduced to fewer agents for real-time systems.
- Heavy language models can be replaced with lightweight or distilled models.
- Can be extended to multilingual and conversational speech.
- Suitable for implementation in assistive communication tools, mobile apps, and voice-based interfaces.

▪ PROBLEM DESCRIPTION

In recent years, voice-based technologies such as automatic speech recognition (ASR), virtual assistants, and voice-enabled communication systems have become an integral part of daily life. However, these systems are primarily trained on fluent speech and perform poorly when processing speech from people who stutter (PWS). Stuttering is a speech disorder characterized by repetitions, prolongations, and blocks, which often cause ASR systems to misrecognize words, distort sentence meaning, or fail completely.

Due to these limitations, people who stutter face significant challenges while using voice-driven applications in academic, professional, and social environments. Existing ASR systems frequently produce inaccurate transcriptions, leading to semantic errors that change the speaker's intended message. This results in exclusion from voice-based services such as online meetings, voice authentication systems, and assistive communication tools.

Current research efforts mainly focus on detecting or labeling stuttering events rather than repairing the distorted output produced by ASR systems. Moreover, direct speech-to-speech conversion for stuttered input is technically complex and unreliable, especially in real-time scenarios. Another major challenge is preserving the original semantic intent of the speaker while removing stuttering-related disfluencies, without introducing additional delays.

Therefore, there is a critical need for an intelligent, real-time system that can accurately interpret stuttered speech, repair semantic distortions caused during transcription, and generate fluent, natural-sounding speech while maintaining the speaker's identity. The absence of such inclusive and accessible speech-processing solutions continues to marginalize people who stutter in a rapidly evolving voice-driven digital society.

▪ PROBLEM STATEMENT

Design and develop a real-time, multi-agent speech processing system that converts stuttered speech into fluent, semantically accurate text and speech output to improve accessibility for people who stutter.

▪ OBJECTIVES

- To design and implement a multi-stage speech processing system that accurately converts stuttered speech into semantically correct text using automatic speech recognition and intelligent text repair.
  (Functional Objective)

- To develop a fluent speech generation module that synthesizes natural, stutter-free audio while preserving the original speaker's voice characteristics.
  (Functional Objective)

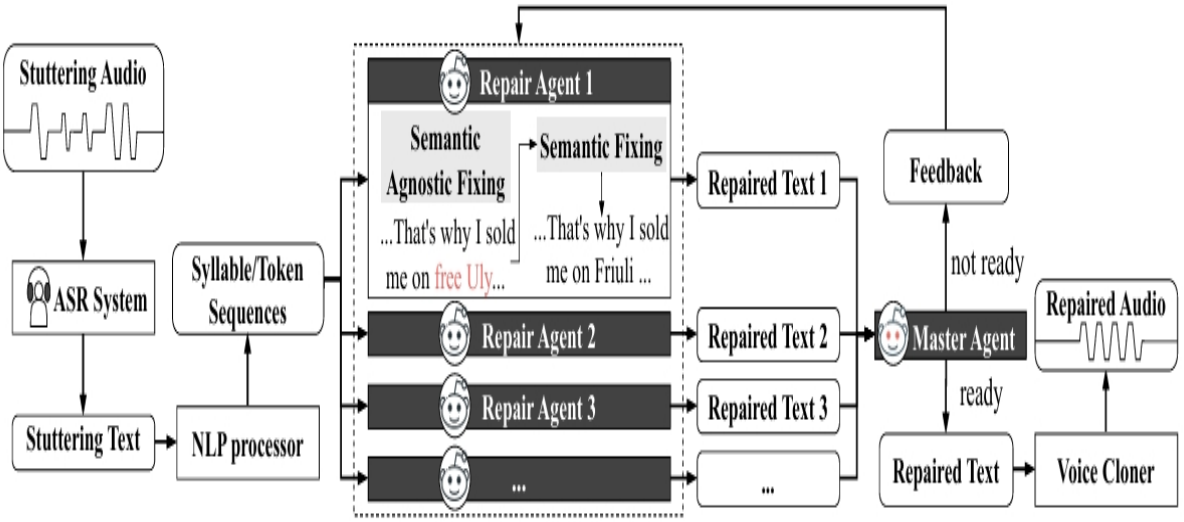
- To evaluate and optimize the system's performance in terms of accuracy, semantic preservation, and real-time latency for effective communication.
  (Performance Objective)

- To incorporate a robust stutter detection and classification mechanism that identifies different types of disfluencies (repetitions, prolongations, blocks) to enable precise downstream processing.

- To ensure the system's usability and adaptability across diverse speakers and acoustic environments, including variations in accent, speaking rate, and background noise.
(Non-Functional Objective)

METHODOLOGY

➢ ARCHITECTURAL DIAGRAM:-



➢ MODULE DESCRIPTION:-

The proposed system follows a multi-stage pipeline architecture consisting of speech recognition, semantic repair using multi-agent collaboration, and fluent speech synthesis. The architectural diagram illustrates how stuttered speech is processed step-by-step to produce semantically accurate and fluent audio output. The design emphasizes modularity, semantic preservation, and real-time performance.

○ Module 1: Speech-to-Text Conversion (ASR Module)-

This module is responsible for converting the stuttered audio input into an initial textual representation. The system accepts stuttering speech as input and processes it using an Automatic Speech Recognition (ASR) engine. Due to repetitions, prolongations, and blocks in stuttered speech, the generated transcript often contains incorrect or duplicated words.

The output of this module serves as a base for further correction and is passed to the NLP processor for tokenization. This module plays a crucial role in capturing the speaker's spoken content while enabling downstream semantic repair.

- ❖ Input: Stuttering audio
- ❖ Output: Stuttered text transcript

o Module 2: Semantic-Aware Multi-Agent Text Repair Module-

The stuttered text obtained from the ASR module is first processed by an NLP processor, which converts the text into token sequences.These sequences are then provided to multiple Agents.

Each Repair Agent independently performs text correction by:

- Removing stuttering-related repetitions and fillers
- Restoring grammatical structure
- Preserving semantic intent

If the repaired text does not meet quality criteria, feedback is sent back to the Repair Agents for refinement.

- ❖ Input: Stuttered text
- ❖ Output: Final repaired and semantically accurate text

o Module 3: Text-to-Speech Conversion (Voice Cloning Module)

Once the repaired text is finalized, it is passed to the Text-to-Speech (TTS) module. This module uses a voice cloning mechanism to generate fluent, natural-sounding speech while preserving the original speaker's vocal characteristics such as pitch and tone.

The system employs a zero-shot voice cloning approach, allowing it to synthesize speech without requiring prior voice samples from the user. This module ensures that the final output audio is free from stuttering and suitable for real-time communication.

- ❖ Input: Repaired text
- ❖ Output: Fluent, stutter-free audio

▪ FACILITIES REQUIRED

- **Personal Computer with Required Specifications-**

A personal computer or laptop with a minimum Intel Core i5 / AMD Ryzen 5 processor, 8 GB RAM (16 GB recommended), and at least 50 GB of free storage is required to support application development, model execution, and testing. A GPU-enabled system (NVIDIA GPU with CUDA support) is recommended for faster speech processing and model inference.

- **Application Development Tools and Frameworks**

- Programming Language: Python
- Development Environment: VS Code / PyCharm / Jupyter Notebook

- Machine Learning Libraries: PyTorch, Transformers, NumPy

- Speech Processing Tools: ASR and Text-to-Speech APIs or open-source frameworks
- Special Software, Libraries, and Datasets

- **Other special hardware/software/dataset requirement-**

- Automatic Speech Recognition Model (e.g., Whisper or equivalent)
- Large Language Model (LLM) API for multi-agent text repair

- Text-to-Speech / Voice Cloning Model (e.g., StyleTTS or similar
- Stuttering Speech Dataset (such as FluencyBank or publicly available speech datasets)

- Internet connectivity for model access, updates, and API usage

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